

\*\*\*\*\*  
\* STAAD.Pro Version 2003 Bld 1001.US \*  
\* Proprietary Program of \*  
\* Research Engineers, Intl. \*  
\* Date= JUL 17, 2009 \*  
\* Time= 19:15:44 \*  
\* USER ID: \*\*\*\*\*  
\*\*\*\*\*

1. STAAD PLANE DXF IMPORT OF 2F7.5K.DXF  
2. START JOB INFORMATION  
3. JOB NAME 2F 7.5K X 22K - DOC. ROOM  
4. JOB CLIENT AEL  
5. ENGINEER NAME RCZ  
6. APPROVED NAME LCK  
7. ENGINEER DATE 15-JUN-09  
8. END JOB INFORMATION  
9. INPUT WIDTH 79  
10. UNIT MMS KN  
11. JOINT COORDINATES  
12. 1 0.420791 0.228998 0; 2 0.420791 3200.23 0; 3 0.420791 6100.23 0  
13. 4 13650.4 0.228998 0; 5 13650.4 3200.23 0; 6 13650.4 6100.23 0  
14. 7 7280.42 0.228998 0; 8 7280.42 3200.23 0; 9 7280.42 6100.23 0  
15. 10 455.421 6205.27 0; 11 1365.42 6415.36 0; 12 2275.42 6625.45 0  
16. 13 3185.42 6835.54 0; 14 4095.42 7045.63 0; 15 5005.42 7255.72 0  
17. 16 5915.42 7465.81 0; 17 6825.42 7675.9 0; 18 7280.42 7570.86 0  
18. 19 8190.42 7360.77 0; 20 9100.42 7150.68 0; 21 10010.4 6940.59 0  
19. 22 10920.4 6730.5 0; 23 11830.4 6520.41 0; 24 12740.4 6310.32 0  
20. 25 455.421 6100.23 0; 26 1365.42 6100.23 0; 27 2275.42 6100.23 0  
21. 28 3185.42 6100.23 0; 29 4095.42 6100.23 0; 30 5005.42 6100.23 0  
22. 31 5915.42 6100.23 0; 32 6825.42 6100.23 0; 33 7280.42 6100.23 0  
23. 34 9100.42 6100.23 0; 35 10010.4 6100.23 0; 36 10920.4 6100.23 0  
24. 37 11830.4 6100.23 0; 38 12740.4 6100.23 0; 39 0.420791 2750.23 0  
25. 40 0.420791 5750.23 0; 41 13650.4 2750.23 0; 42 13650.4 5750.23 0  
26. 43 7280.42 2750.23 0; 44 500.421 3200.23 0; 45 6780.42 3200.23 0  
27. 46 7780.42 3200.23 0; 47 13150.4 3200.23 0  
28. MEMBER INCIDENCES  
29. 1 1 39; 2 2 40; 3 4 41; 4 5 42; 5 7 43; 6 8 9; 7 3 10; 8 10 11; 9 11 12  
30. 10 12 13; 11 13 14; 12 14 15; 13 15 16; 14 16 17; 15 17 18; 16 18 19; 17 19 20  
31. 18 20 21; 19 21 22; 20 22 23; 21 23 24; 22 24 25; 23 25 26; 24 26 27  
32. 26 27 28; 27 28 29; 28 29 30; 29 30 31; 30 31 32; 31 32 33; 32 33 34  
33. 34 34 35; 35 35 36; 36 36 37; 37 37 38; 38 38 39; 39 17 32; 40 25 10; 41 10 26  
34. 42 26 11; 43 11 27; 44 27 32; 45 12 28; 46 28 13; 47 13 29; 48 29 14; 49 14 30  
35. 50 30 15; 51 15 31; 52 31 16; 53 16 32; 54 32 18; 55 18 9; 56 9 19; 57 19 33  
36. 58 33 20; 59 20 34; 60 34 21; 61 21 35; 62 35 22; 63 22 36; 64 36 23; 65 23 37  
37. 66 37 24; 67 24 38; 68 2 44; 69 8 46; 70 39 2; 71 40 3; 72 41 5; 73 42 6  
38. 74 43 8; 75 44 45; 76 45 8; 77 46 47; 78 47 5; 79 39 44; 80 43 45; 81 43 46  
39. 82 41 47; 83 40 25; 84 42 38  
40. DEFINE MATERIAL START  
41. ISOTROPIC STEEL

DXF IMPORT OF 2F7.5K.DXF

42. E 205  
43. POISSON 0.3  
44. DENSITY 7.881952E-008  
45. ALPHA 6.5E-006  
46. DAMP 0.03  
47. END DEFINE MATERIAL  
48. CONSTANTS  
49. MATERIAL STEEL ALL  
50. START USER TABLE  
51. TABLE 1  
52. UNIT MMS KN  
53. GENERAL  
54. C38X38  
55. 252 38 2.3 38 2.3 38072 62913 100000 1557 3311 87 175 1557 3311 100000 28  
56. TABLE 2  
57. UNIT MMS KN  
58. GENERAL  
59. 2C75X45  
60. 1098 75 6 90 3 938192 638334 100000 26218 14185 450 270 26218 14185 100000 75  
61. TABLE 3  
62. UNIT MMS KN  
63. GENERAL  
64. C100X50  
65. 653 50 3 100 3 229000 990000 1E+006 7270 19800 300 343 11600 23600 1E+006 48  
66. TABLE 4  
67. UNIT MMS KN  
68. GENERAL  
69. C75X45  
70. 518 45 3 75 3 138000 449000 100000 4940 12000 270 315 4940 12000 100000 40  
71. TABLE 5  
72. UNIT MMS KN  
73. TUBE  
74. SHS100X3-CF  
75. 1140 100 100 3 1.77E+006 1.77E+006 2.79E+006 600 400  
76. TABLE 6  
77. UNIT MMS KN  
78. TUBE  
79. SHS100X4-CF  
80. 1490 100 100 4 2.26E+006 2.26E+006 3.62E+006 800 533  
81. TABLE 7  
82. UNIT MMS KN  
83. TUBE  
84. SHS100X6-CF  
85. 2160 100 100 6 3.11E+006 3.11E+006 5.14E+006 1200 800  
86. TABLE 8  
87. UNIT MMS KN  
88. TUBE  
89. SHS150X4.5CF  
90. 2570 150 150 4.5 8.96E+006 8.96E+006 1.41E+007 1350 900  
91. TABLE 9  
92. UNIT MMS KN  
93. TUBE  
94. SHS75X3-CF  
95. 841 75 75 3 716000 716000 1.15E+006 450 300  
96. TABLE 10  
97. UNIT MMS KN

DXF IMPORT OF 2F7.5K.DXF

98. TUBE  
 99. SHS50X3-CF  
 100. 541 50 50 3 195000 195000 321000 300 200  
 101. TABLE 11  
 102. UNIT MMS KN  
 103. TUBE  
 104. SHS38X3-CF  
 105. 297 38 38 3 78500 78500 133000 228 152  
 106. TABLE 12  
 107. UNIT MMS KN  
 108. GENERAL  
 109. 2C300X9X3.0  
 110. 3358 300 6 196 3 4.416E+007 6.5986E+006 100000 294408 68736 2136 1800 -  
 111. 9.29441E+006 68736 100000 300  
 112. TABLE 13  
 113. UNIT MMS KN  
 114. TUBE  
 115. SHS125X4.5-CF  
 116. 2120 125 125 4.5 5.06E+006 5.06E+006 8.04E+006 1125 750  
 117. TABLE 14  
 118. UNIT MMS KN  
 119. TUBE  
 120. RSH150X100X4.5-CF  
 121. 2120 150 110 4.5 6.58E+006 3.52E+006 7.36E+006 900 900  
 122. TABLE 15  
 123. UNIT MMS KN  
 124. TUBE  
 125. SHS75X5-CF  
 126. 1340 75 75 5 1.06E+006 1.08E+006 1.77E+006 750 500  
 127. TABLE 16  
 128. UNIT MMS KN  
 129. TUBE  
 130. SHS75X6-CF  
 131. 1560 75 75 6 1.2E+006 1.2E+006 1.97105E+006 900 600  
 132. TABLE 17  
 133. UNIT MMS KN  
 134. TUBE  
 135. SHS75X4.5-CF  
 136. 970 75 75 4.5 986000 986000 1.63E+006 675 450  
 137. END  
 138. MEMBER PROPERTY AMERICAN  
 139. 2 TO 4 71 TO 73 UPTABLE 2 2C75X45  
 140. 5 6 74 UPTABLE 8 SHS150X4.5CF  
 141. 68 69 75 TO 78 UPTABLE 12 2C300X9X3.0  
 142. 7 TO 39 UPTABLE 4 C75X45  
 143. 40 TO 67 UPTABLE 11 SHS38X3-CF  
 144. 83 84 UPTABLE 10 SHS50X3-CF  
 145. UNIT METER KN  
 146. MEMBER PROPERTY AMERICAN  
 147. 79 TO 82 UPTABLE 16 SHS75X6-CF  
 148. MEMBER PROPERTY AMERICAN  
 149. 1 70 UPTABLE 17 SHS75X4.5-CF  
 150. UNIT MMS KN  
 151. SUPPORTS  
 152. 1 4 7 PINNED  
 153. MEMBER TRUSS

DXF IMPORT OF 2F7.5K.DXF

154. 39 TO 67  
 155. MEMBER RELEASE  
 156. 7 23 68 69 START MX MY MZ  
 157. 38 76 78 END MX MY MZ  
 158. 79 82 START MP 0.95  
 159. UNIT METER KN  
 160. LOAD 1 S/W  
 161. SELFWEIGHT Y -1  
 162. LOAD 2 DL  
 163. MEMBER LOAD  
 164. 7 TO 22 UNI GY -1.27  
 165. 68 69 75 TO 78 UNI GY -0.76  
 166. JOINT LOAD  
 167. 8 FY -12.4  
 168. 8 FY -4.52  
 169. 9 FY -9.55  
 170. LOAD 3 LL  
 171. MEMBER LOAD  
 172. 7 TO 22 UNI GY -1.82  
 173. 69 77 78 UNI GY -4.55  
 174. JOINT LOAD  
 175. 8 FY -58.96  
 176. 8 FY -23.81  
 177. 9 FY -8.75  
 178. MEMBER LOAD  
 179. 68 75 76 UNI GY -9.1  
 180. LOAD 4 WL  
 181. MEMBER LOAD  
 182. 7 TO 22 UNI GY 1  
 183. JOINT LOAD  
 184. 9 FY 5  
 185. LOAD COMB 6 ULTIMATE S/W-DL+LL  
 186. 1 1.4 2 1.4 3 1.6  
 187. LOAD COMB 7 ULTIMATE DL+WL  
 188. 1 1.0 2 1.0 4 1.4  
 189. LOAD COMB 9 SERVICE DL+LL FOR DEFLECTION  
 190. 2 1.0 3 1.0  
 191. LOAD COMB 10 SERVICE S/W-DL+LL FOR FOUNDATION DESIGN  
 192. 1 1.0 2 1.0 3 1.0  
 193. PERFORM ANALYSIS

## PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER-ELEMENTS/SUPPORTS = 47/ 84/ 3  
 ORIGINAL/TOTAL BAND-WIDTH= 42/ 5/ 18 DOF  
 TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 135  
 SIZE OF STIFFNESS MATRIX = 3 DOUBLE Kilo-WORDS  
 REORD/AVAIL. DISK SPACE = 12.2/ 32685.3 MB, EXMEM = 251.9 MB